

Nathan Leclercq

Data Engineer - MLOps

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Profile

Data Engineer & MLOps at DataKhi for 3 years (apprenticeship, then permanent contract since Sept. 2025). I design and operate end-to-end data platforms — ingestion, ETL pipelines, warehousing, deployment, monitoring — and cover the full ML lifecycle: data preparation, feature engineering, model design and training, then production deployment. Equally comfortable with **open-source** stacks I deploy and run myself (PostgreSQL, Airflow, Kubernetes, Ansible) and with managed cloud (Microsoft Fabric, Azure). Background in mathematics, Master's in Machine Learning (Lille).

Experience

Nyukom — telecom data platform

DataKhi · Permanent · Data Engineer

Oct. 2025 – present

- End-to-end telecom data platform for multi-client steering: from field collection (3CX, Centreon API) to Power BI reports, through an object data lake and a **PostgreSQL** star-schema warehouse
- Infrastructure deployed and operated autonomously on **Kubernetes** (K3s), orchestrated by **Apache Airflow** and provisioned with **Ansible**: idempotent, partitioned pipelines, historical backfill and error recovery
- Migrated object storage from **MinIO** to **Garage** to harden sensitive workloads with zero downtime (D-1 mirror maintained during acceptance testing)

Company reference dataset & B2B prospecting engine

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2025 – present

- National company reference dataset for sales teams: **ETL** pipeline feeding a **PostgreSQL** warehouse of nearly 15M companies and 23M establishments (INSEE), enriched with financial ratios (BCE/INPI) and director data via public APIs
- Robust, replayable ingestion: full mode (atomic swap preserving enrichments) and incremental mode (UPSERT), error-tolerant, deployed as a **Kubernetes CronJob**
- Automated prospecting chain on top of the dataset: prospect targeting, website resolution, verified contact lookup, and personalized email generation via **LLM** (Mistral)

Object storage benchmark (Garage vs MinIO)

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2025 – 2026

- **Garage vs MinIO** benchmark (multi-node cluster) to decide on object storage: **DuckDB** read throughput, S3 writes, resilience (node failure + recovery)
- Reproducible, automated test bench (multi-node Docker Compose + deployment scripts) — technical foundation for the **DevLille 2026** talk

Hall U Need data infrastructure

DataKhi · Permanent · Data Engineer

2023 – present

- Run and evolution of a production data platform for a restaurant company: **Microsoft Fabric** (warehouse + notebooks), **Azure Data Factory** (orchestration), custom Playwright scraping pipelines for non-Azure sources
- Day-to-day: incident handling, cloud cost control, integration of new business sources (POS, reservations, weather, HR, events)

Hall U Need football forecasting

DataKhi · Permanent · ML Engineer

2023 – present

- Industrialized a football forecasting system (28 **XGBoost** quantile-regression models) to size staffing and stock: rich feature engineering (weather, calendar, public holidays, reservations), custom Huber loss, confidence-interval calibration
- Production deployment in **Microsoft Fabric**: **Azure Data Factory** pipeline orchestrating collection → periodic training → daily prediction (90-day horizon), models versioned (date + MAE) on OneLake
- Long-term reliability: data-quality checks with email alerting on anomalies, non-regression tests (Pearson ≥ 0.999), error tracking (MAE, interval coverage) and retrospective predicted-vs-actual analysis

Tossée — eco-responsible fashion aggregator

- DataKhi · Apprenticeship · Data / MLOps Engineer2023 – 2025
- Complete data ecosystem for an eco-responsible fashion aggregator: batch chain collection → normalization → ML enrichment → API, over a 67-brand catalogue
 - Product environmental-impact score (24 PEF indicators via the **Ecobalyse** API): material mapping, caching and reliability-based deduplication to limit calls
 - Large-scale collection (Scrapy, Playwright, in-house YAML rules engine) and a **PostgreSQL/pgvector** warehouse for semantic search, exposed via a **FastAPI** API
 - Pipelines deployed as **Kubernetes** (K3s) workers orchestrated by a message queue, hybrid on-premise / Azure deployment

Power BI versioning — internal project

- DataKhi · Internship · Full-Stack Developer2023
- Power BI versioning system: C++ backend (report diffing), React frontend, Electron distribution

Technical Skills

Data Engineering	PostgreSQL · SQL · Apache Airflow · Spark · ETL/ELT pipelines · star schema · partitioning · idempotency/backfill · PyArrow/Parquet · DuckDB · Microsoft Fabric · Azure Data Factory
Infra & Storage	Linux/Bash · Kubernetes (K3s) · Docker · Ansible & Terraform (IaC, vault) · MinIO/Garage (S3) · Prometheus/Grafana/Loki · CI/CD (Forgejo, GitLab) · Azure
Programming	Python (Pandas, PyArrow, FastAPI) · SQL · Bash · Julia · basics of Go, Rust
ML / MLOps	XGBoost · feature engineering · pgvector/embeddings · RAG (MCP, Ollama) · MLflow · CamemBERT
Languages	French (native) · English (professional, TOEIC 885)

Education

2023 – 2025	Master's in Machine Learning, University of Lille	Lille
2020 – 2023	BSc Computer Science, University of Lille	Lille
2017 – 2020	Mathematics (undergraduate, 3 years), University of Lille	Lille

Talks & Publications

Speaker — DevLille 2026: “Take back control of your data platform from the American giants” — deploying a data platform in private cloud (MinIO/Garage, PostgreSQL, DuckDB, K3s).

Active tech watch — attendee at several tech conferences: Cloud Nord 2024 and DevLille 2025.

Blog — 8 technical articles on datakhi.fr (sovereign data platform, object storage, open-source migration, Airflow → Kestra, ML) + nathan.leclercq.spacesheep.ovh (Homelab and Book Reco series).